

Perceptual Grouping in Humans and Machines: SAM cannot do contour integration

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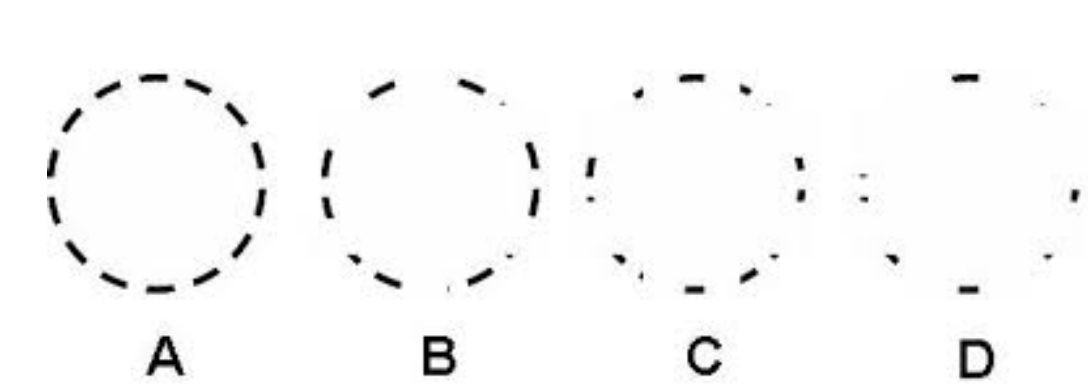
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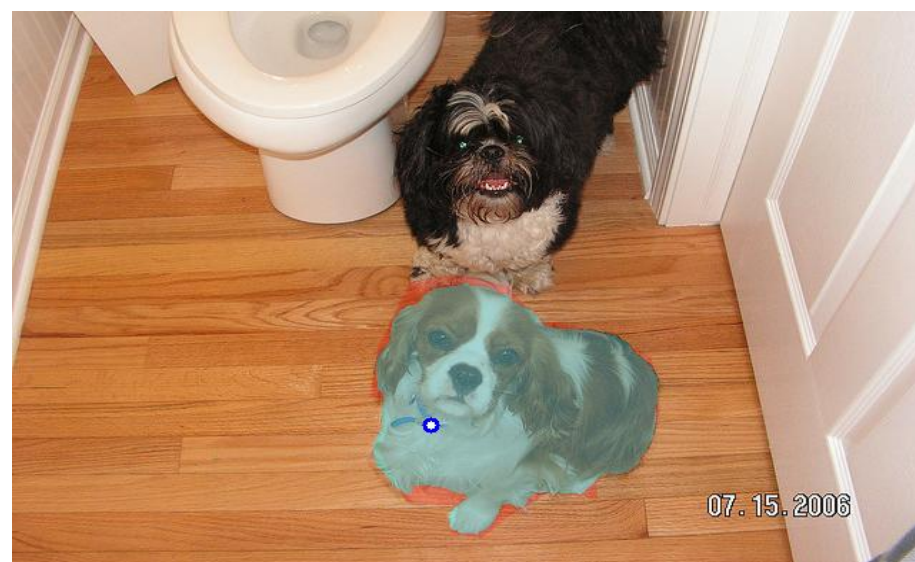
Introduction & Motivation

- Gestalt Principles¹:** Humans perceive “closure” in fragmented contours; computer vision models often fail without semantic cues.
- Segment Anything Model (SAM)^{2,3}:** modern computer vision model for image segmentation
- Goal:** Does SAM replicate human-level Gestalt grouping?

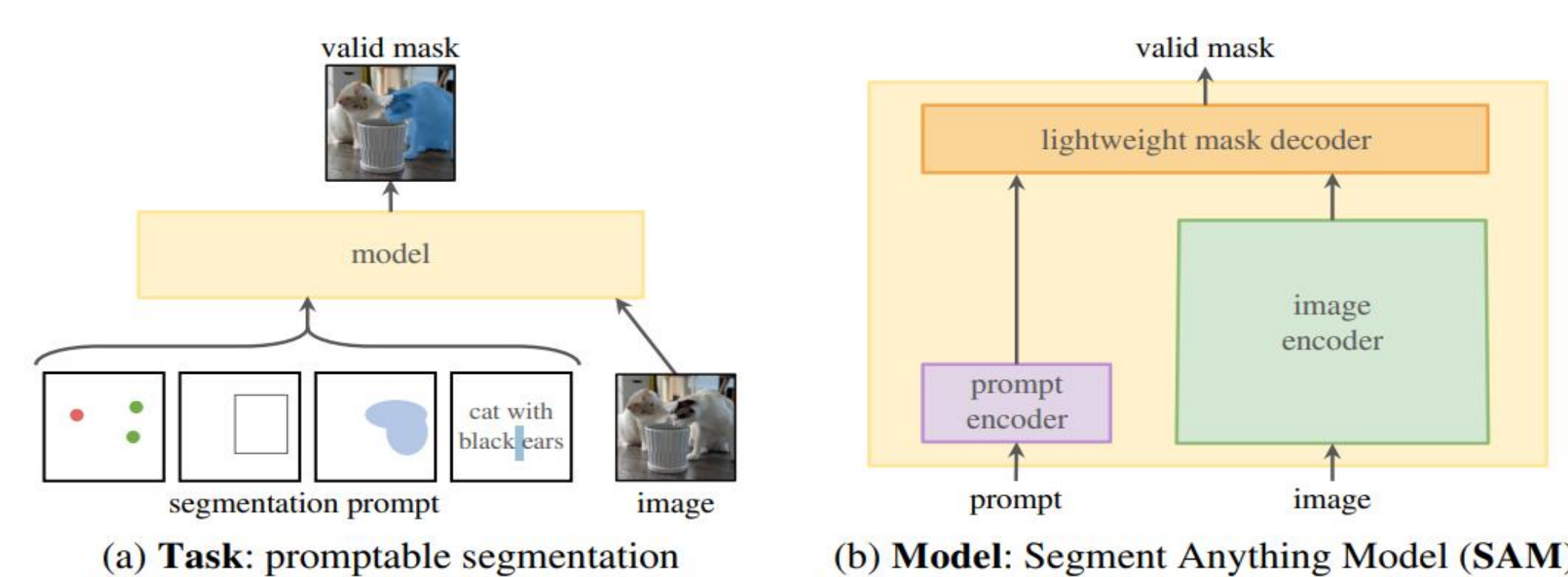
Gestalt Closure



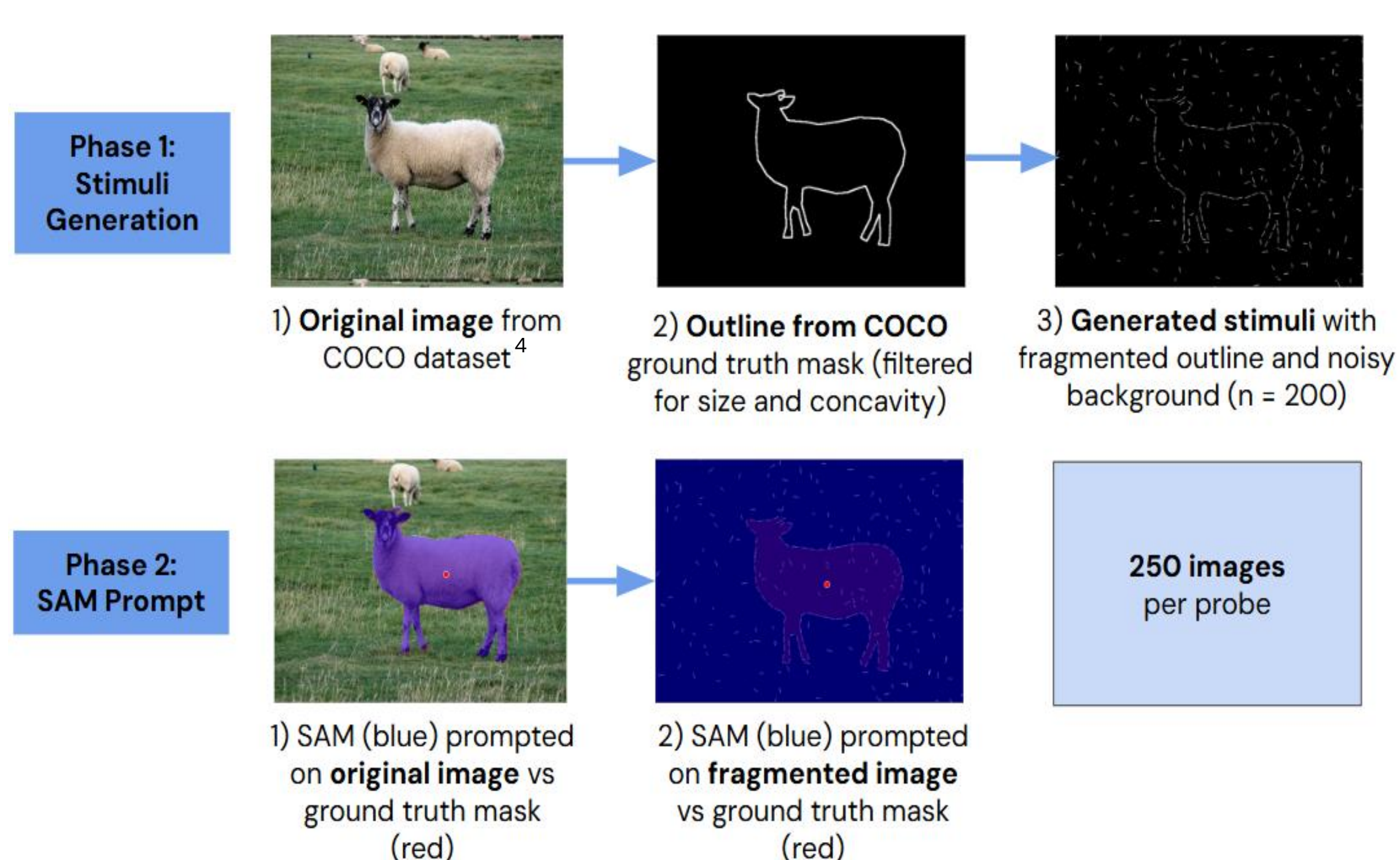
SAM Prompt Example



SAM Architecture

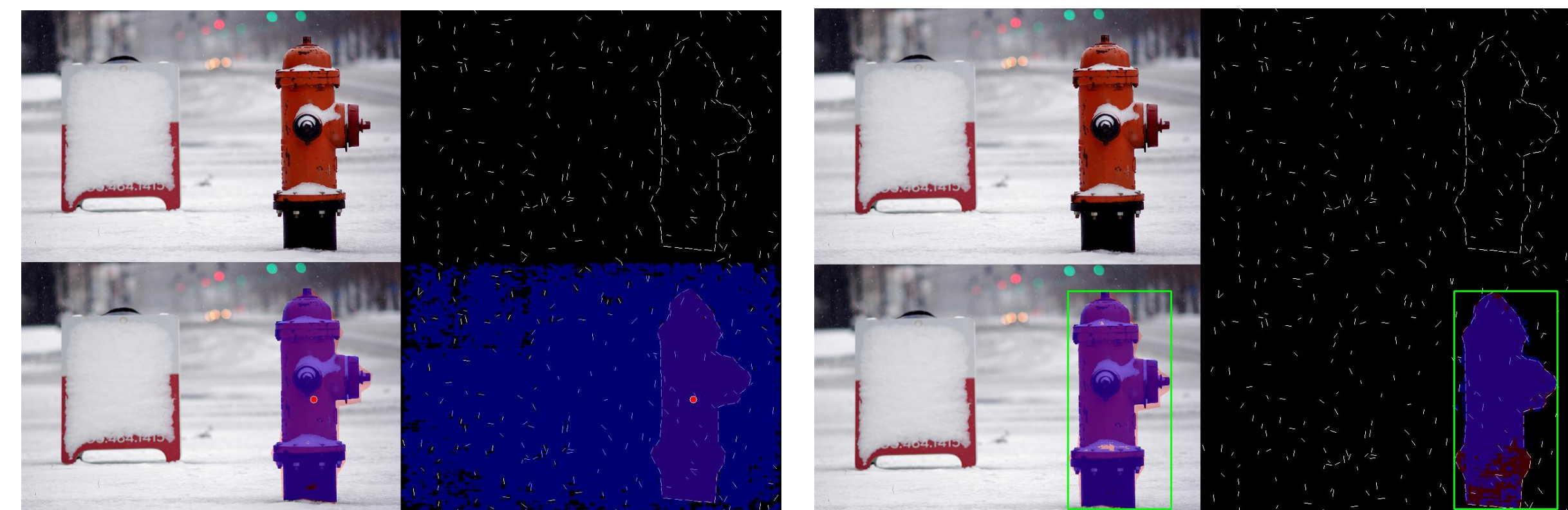


Methods



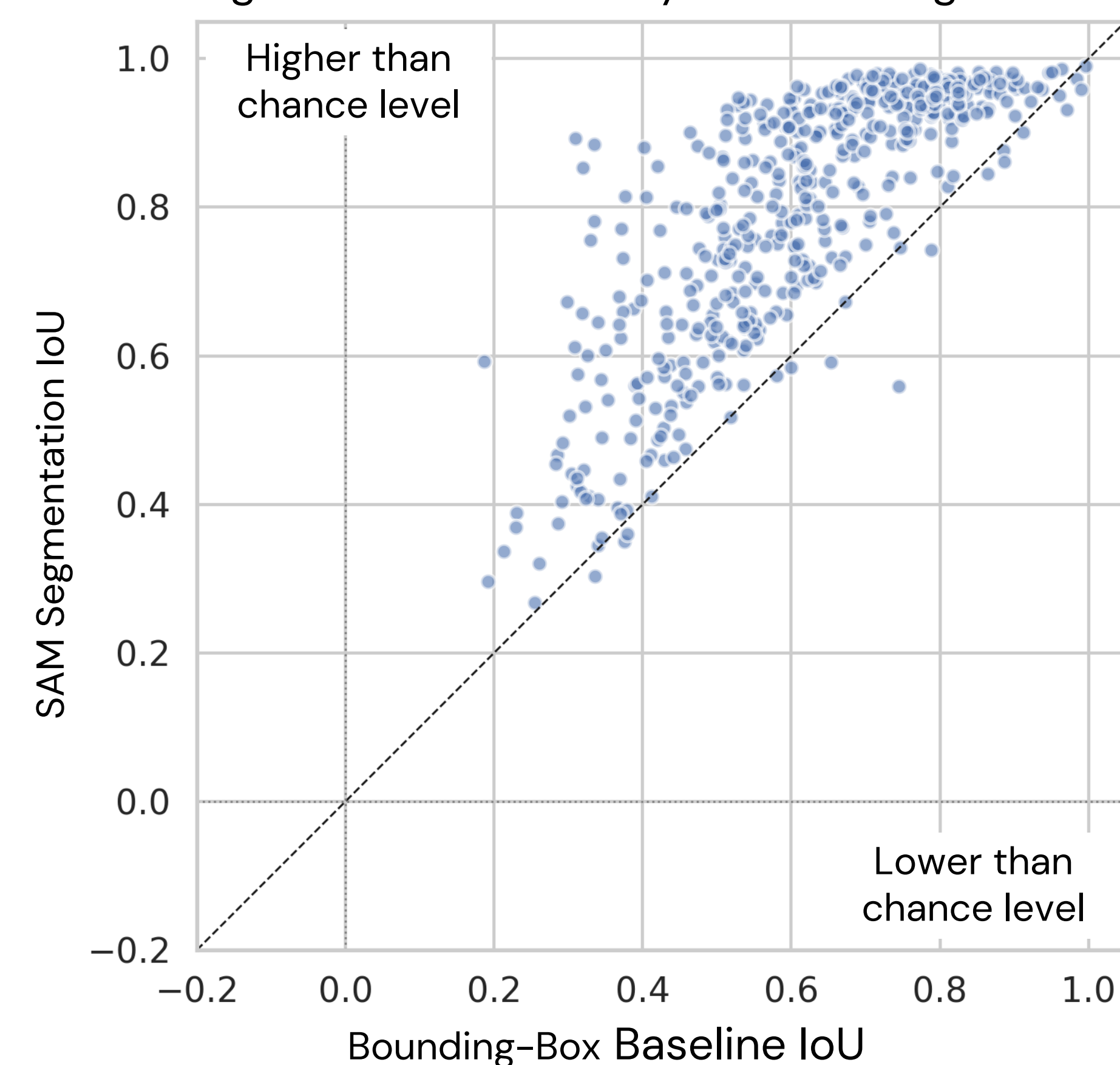
Removing more salient cues (i.e. texture and color) forces reliance on global integration and perceptual grouping

Prompting with Bounding Boxes



In comparison to other prompting strategies, SAM performs significantly better given bounding boxes

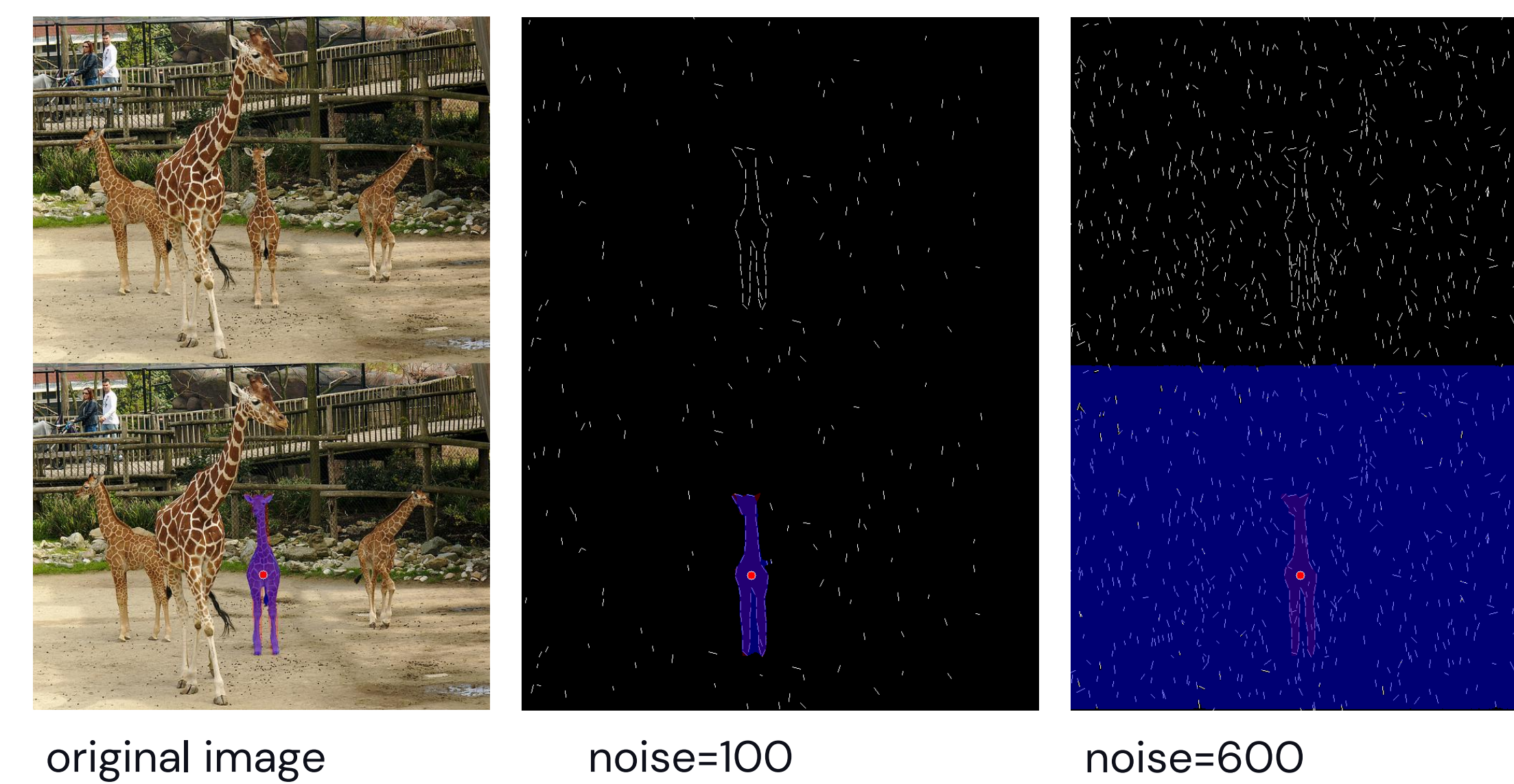
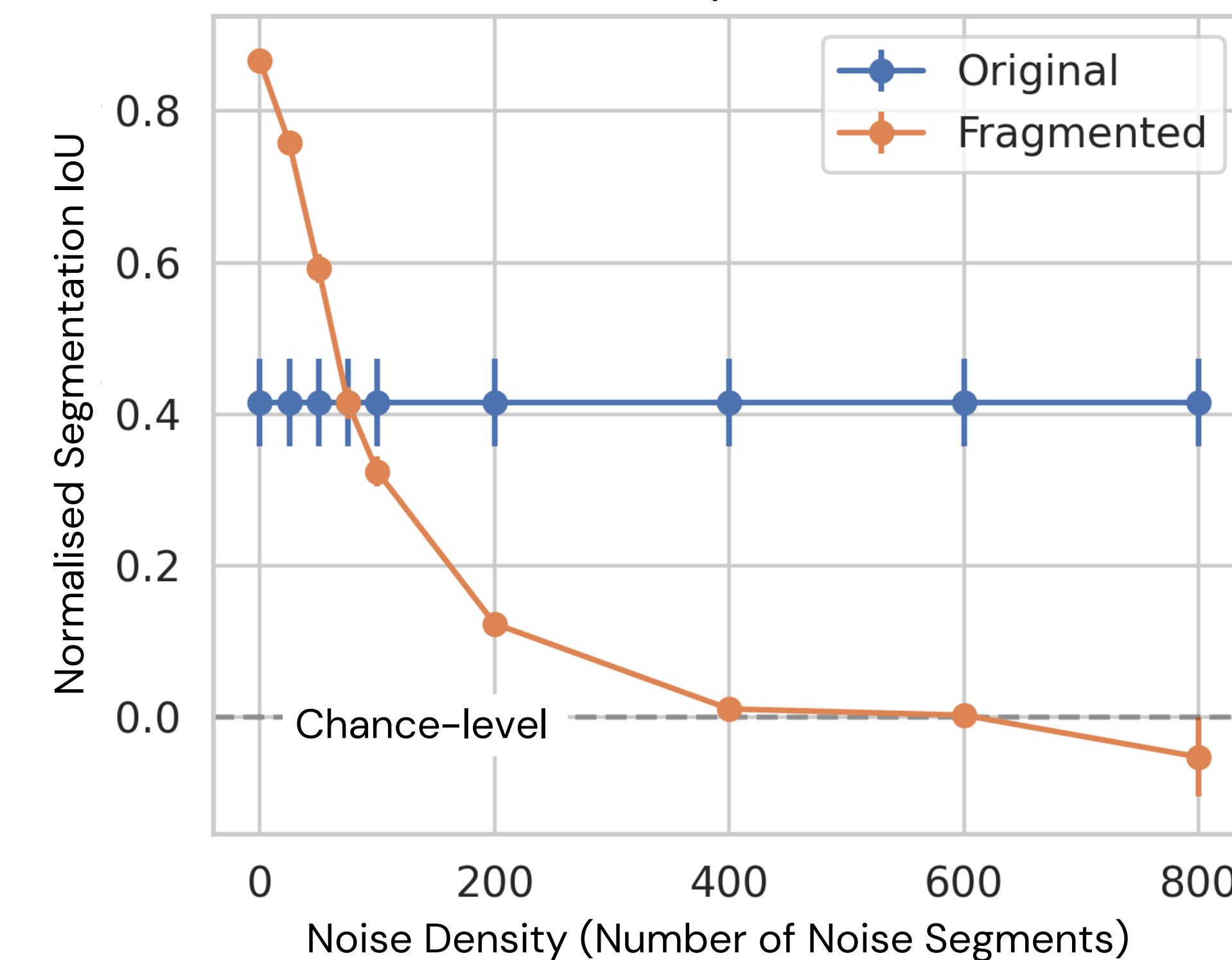
SAM Segmentation Accuracy vs. Bounding-Box Baseline



How much information do we have to give SAM for it to consistently segment fragmented contours?

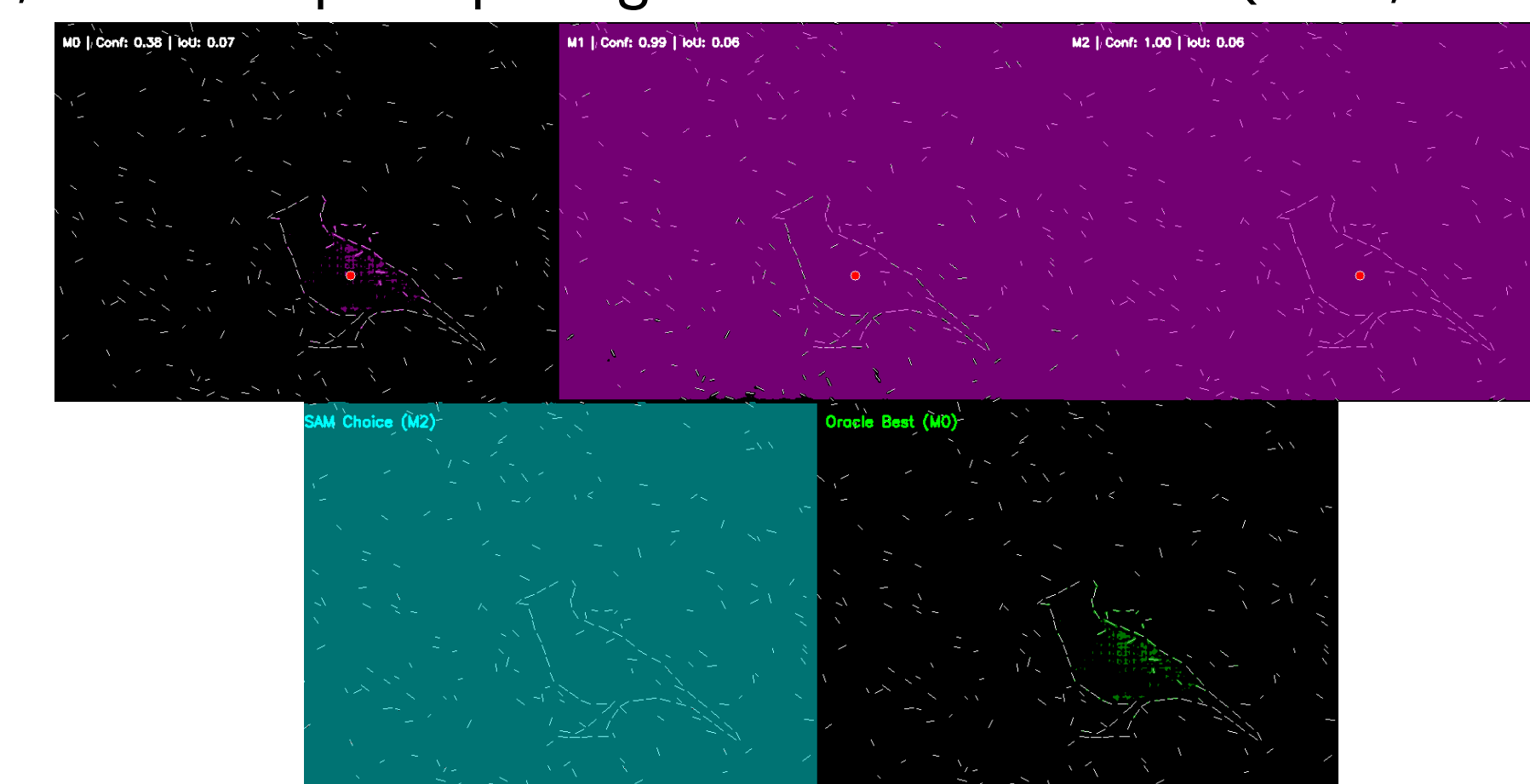
SAM performance collapses with increasing background noise density

Effect of Noise Density on SAM Performance

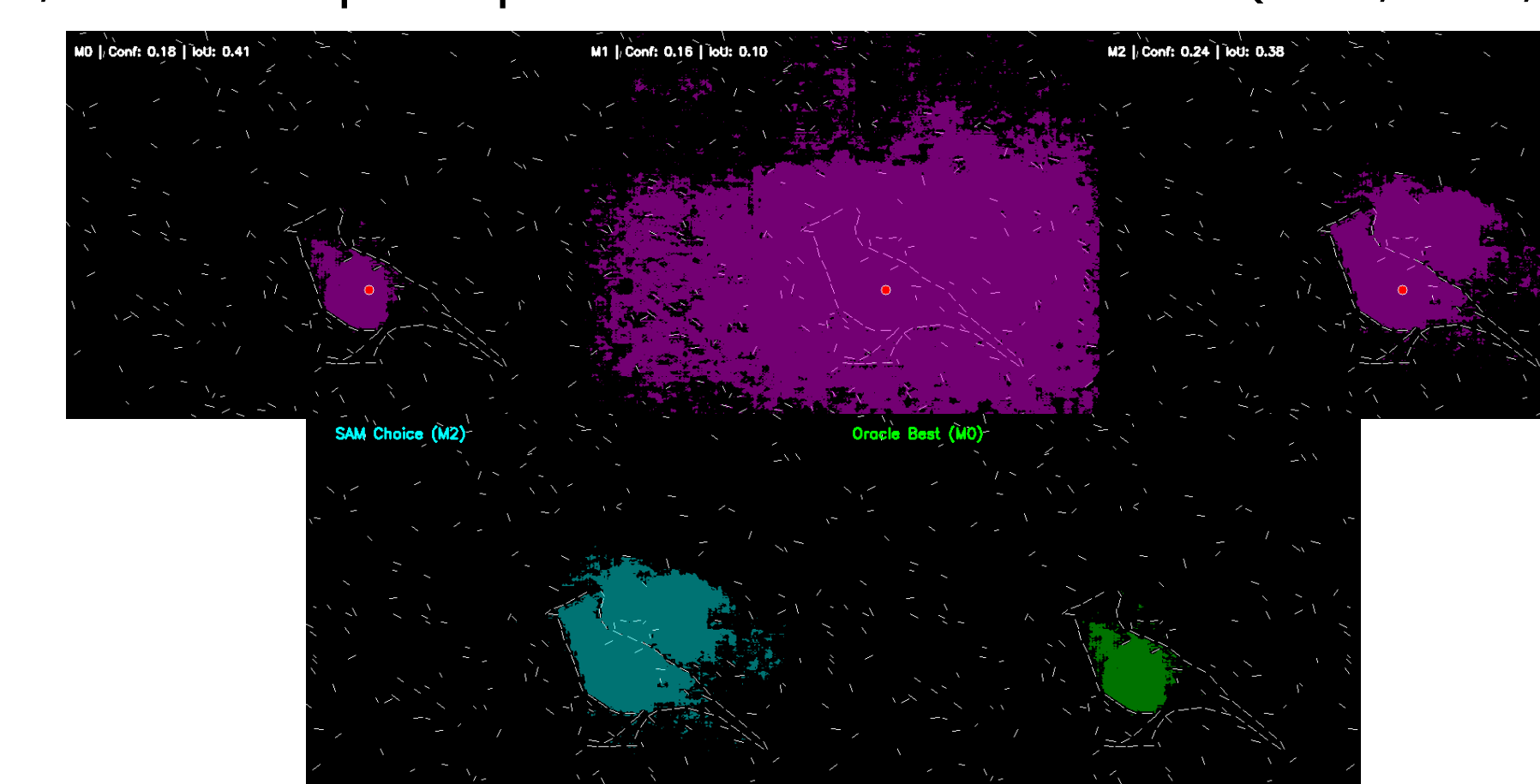


Confidence Levels & Mask Choice

SAM1, centroid prompt: high confidence levels (0.38, 0.99, 1.00)



SAM3, centroid prompt: low confidence levels (0.41, 0.15, 0.24)

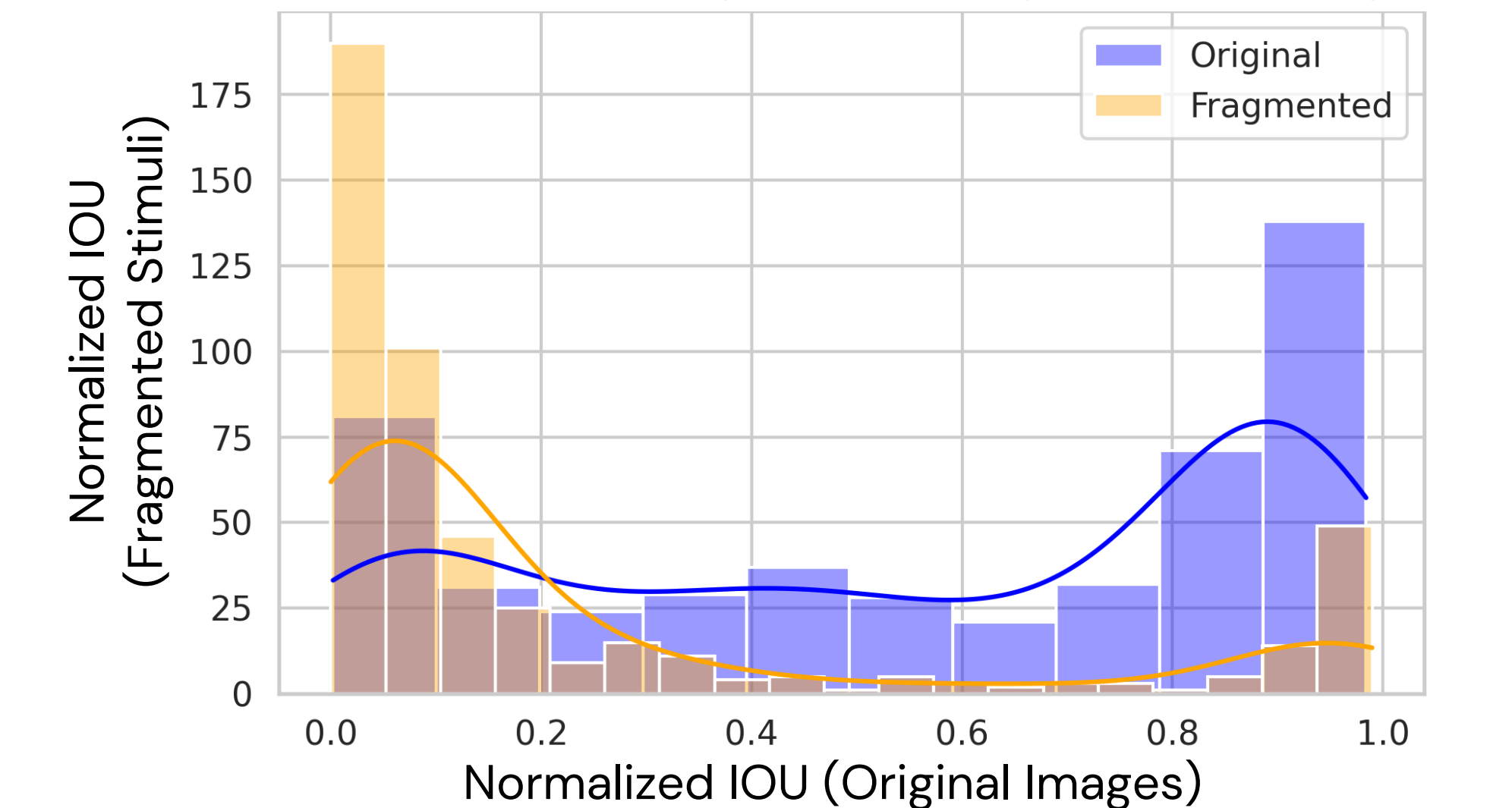


There is often a discrepancy between the optimal mask choice and the choice that SAM makes.

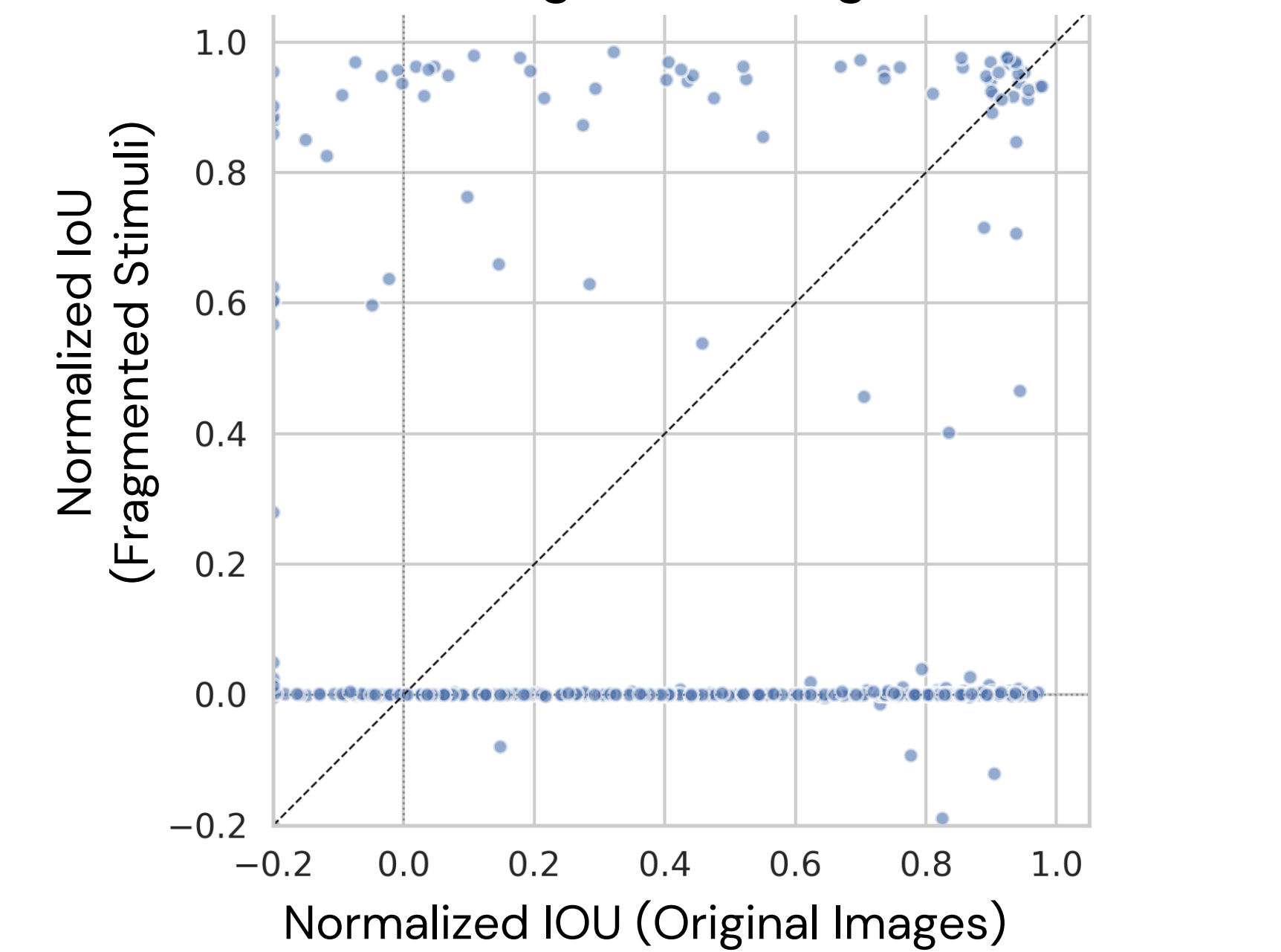
Conclusion

- SAM fails at global perceptual grouping when stripped of salient semantic and texture cues.
- High-level semantic training does not automatically grant a model the mid-level vision capabilities required for Gestalt grouping.

IoU Distribution: Original vs. Fragmented Images

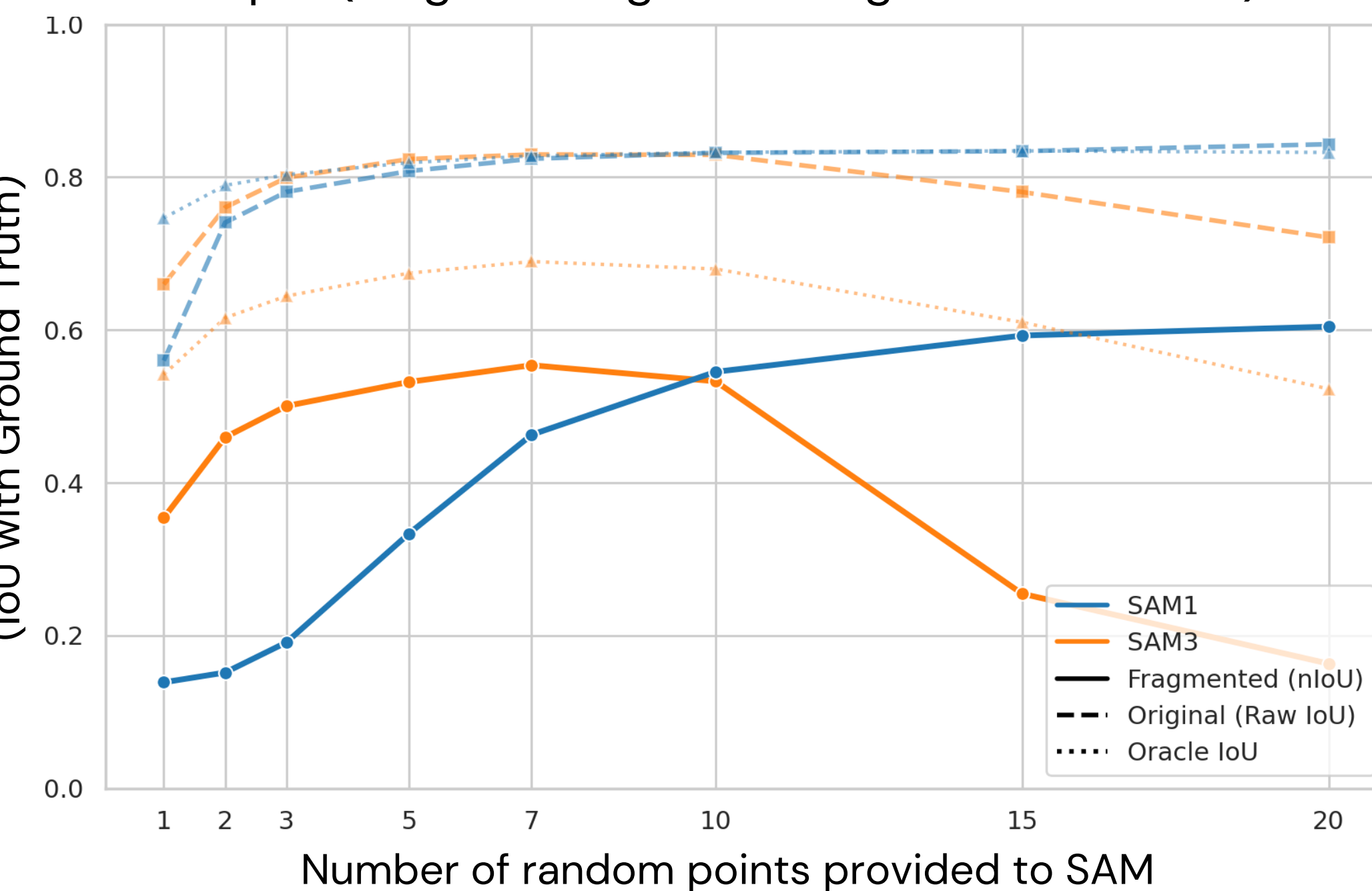


Normalized IoU: Original vs. Fragmented Stimuli

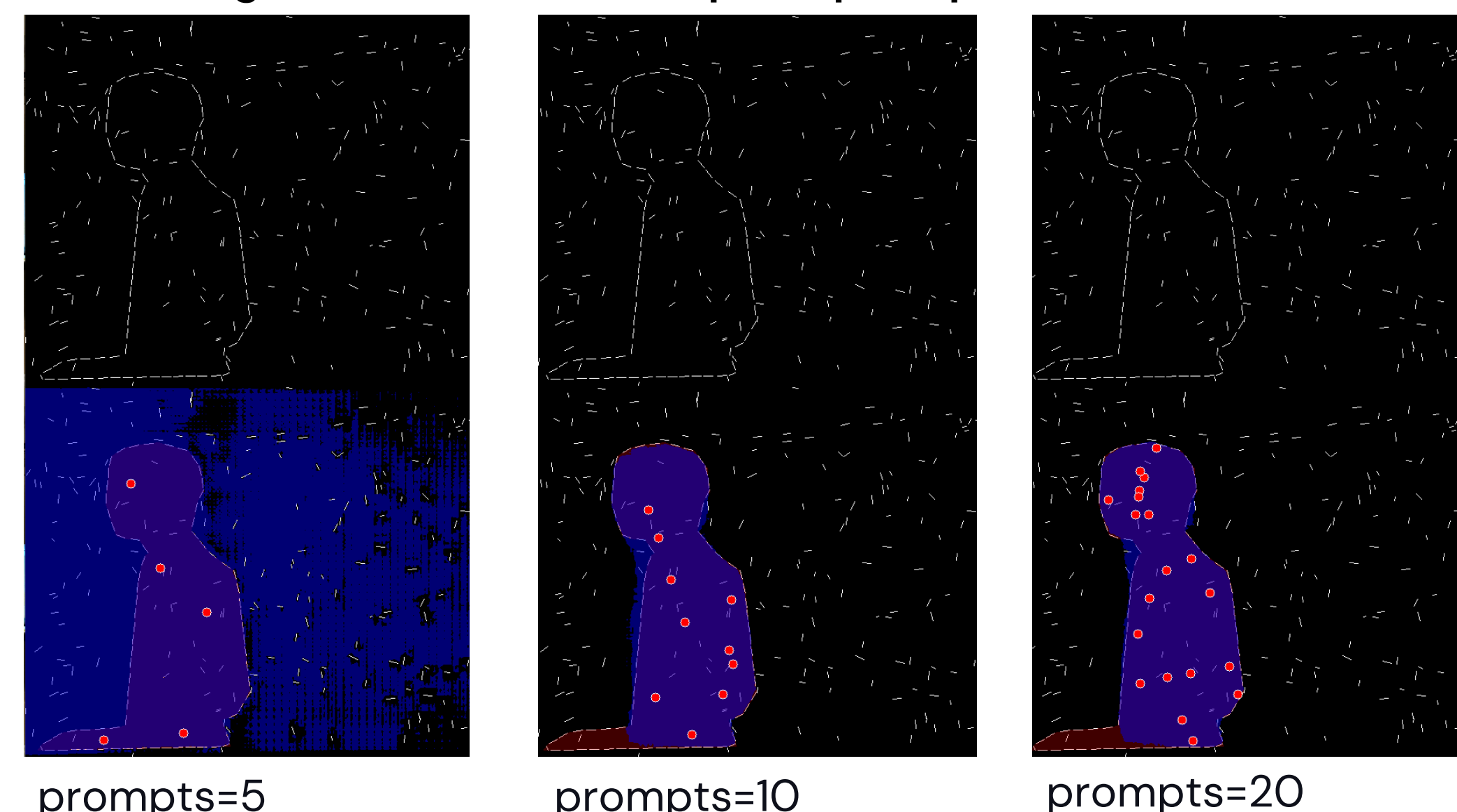


Multi-Prompt Sweep

SAM Segmentation Accuracy vs. Number of Random Point Prompts (Original Images vs. Fragmented Stimuli)



Examples of SAM segmentation with increasing numbers of random point prompts



Discussion & Future Directions

- Selection Gap:** SAM often generates the correct mask (MO–M2) but assigns it low confidence, can we teach it to do better?
- Information Dependency:** High reliance on bounding boxes suggests SAM lacks an internal grouping mechanism and relies more on external features.
- Future Work:** Integrating mid-level vision principles (closure, continuity) into a transformer model via self-attention to the actual boundaries of objects.

References

- [1] Wolfe, J. M., Kluender, K. R., & Levi, D. M. (2020). *Sensation & Perception* (4th ed., Chapter 4). Oxford University Press.
- [2] Kirillov, A., Mintun, E., Ravi, N., et al. (2023). *Segment Anything*. Proceedings of the IEEE/CVF International Conference on Computer Vision. <https://arxiv.org/abs/2304.02643>
- [3] Carion, N., Gustafson, L., Hu, et al. (2025, November 20). *Sam 3: Segment anything with concepts*. <https://arxiv.org/abs/2511.16719>
- [4] Lin, T.-Y., Maire, M., Belongie, S., et al. (2015, February 21). *Microsoft Coco: Common Objects in Context*. <https://arxiv.org/abs/1405.0312>